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Simulation of laminar flow through eccentric annuli using isogeometric variational multiscale method †

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Abstract

We present an application of the Residual-based variational multiscale (RBVMS) methodology to the computation of laminar eccentric annular pipe flow with eccentricities of 0.5 and 1. Isogeometric analysis is utilized for higher order approximation of the geometry and solution using Non-uniform rational B-splines (NURBS) functions. The ability of NURBS exactly representing curved circular geometries makes NURBS-based isogeometric analysis attractive for the application to the flow through the eccentric annuli. By using the exact representation of circular boundary, the limiting case of the eccentricity, i.e. the inner circular wall actually touches the outer circular wall, is successfully demonstrated.

Keywords: Eccentric annular pipes; Incompressible Navier-Stokes equations; Isogeometric analysis; NURBS

1. Introduction

There is a large scientific literature on the flows in an annular geometry. The annular geometry is also widely used in industrial applications: in heat exchangers, food processing, fluidized beds, pulp screening, extrusion, as well as in the domain of oil well construction which forms the motivation for this work. The developing laminar flow and heat transfer in the limiting case of concentric annular passages have been investigated by Heaton et al. [1] Sparrow and Lin [2], Coney and El-Shaarawi [3] and El-Shaarawi and Sarhan [4]. For the eccentric annulus, most of the investigations available in the literature deals with rather simplified cases by assuming potential, slug or fully developed flows. Heyda [5] determined the Green's function in bipolar coordinates for a potential flow and used his potential solution to obtain a solution for the momentum equation. Snyder [6] and Snyder and Goldstein [7] utilized the work of El-Saden [8] to obtain solution for the energy equation describing the slug flow heat transfer under different boundary conditions and the momentum equation describing the fully developed forced laminar flow, respectively. The latter problem was also solved numerically, using bipolar coordinates, by Redberger and Charles [9]. Using cylindrical coordinate, Cheng and Hwang [10] and Trombetta [11] obtained approximate solution for the energy equation under different boundary conditions in the case of fully developed laminar forced flow in eccentric annuli. Using bipolar coordinates Guckes [12] numerically investigated the fully developed laminar flow of non-Newtonian fluids in eccentric annuli. Suzuki et al. [13] and Sathymurthy et al. [14] presented numerical results for the fully developed forced and mixed convection, respectively, in eccentric annuli. Recently, Chen et al. [15] have solved laminar flow problem for the case of eccentric annuli using analytical and semi-analytical Green's function approaches. solved by using analytical and semi-analytical approaches.

The Variational multiscale (VMS) modeling methodology [16] provides a theoretical framework for general multiscale problems in computational mechanics by separating the scales of interest in a predetermined number of groups, usually two, namely, coarse (Resolved) and fine (Unresolved) scales. VMS can be considered as a technique to take into account the effect of neglected unresolved fine scales on the coarse scales. The variational multiscale method was originally proposed as a theoretical justification of established stabilized methods, but has become a platform for the development of new computational technologies (see Ref. [17] for its application to turbulence modeling and simulation of complex flows).

The incompressible Navier-Stokes equations are the mathematical model for both laminar and turbulent flow. In this work we employ the residual-based VMS modeling approach recently proposed in Ref. [17]. The modeling paradigm encompasses the VMS theory of turbulence [16, 18] and the

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numerical experience of stabilized methods [16, 18, 19] that are residual-based. VMS is not based on a concept originated from flow physics, but rather than a concept from the mathematical structure of the Navier-Stokes equations and a multiscale decomposition of the function space of its admissible solutions. In this sense, VMS is a methodology for solving for all regimes of the flows governed by the Navier-Stokes equations. The flow can be laminar, turbulent, or some combination of the two in space and time. This observation has been made previously, and the present work represents an opportunity to test it computationally on concentric annular channel flows.

VMS modeling of flows entails an a priori additive scale separation that yields two equations which govern the dynamics of the coarse and fine scales. We identify the coarse scales with those resolved (Captured) by the computational mesh, while fine unresolved (i.e.,subgrid) scales are not properly captured by the mesh, and their effect on the coarse scales needs to be accounted for. This is done by approximating the problems governing the unresolved scales with local problems that are linearized about the coarse scales and solved utilizing the concept of the fine-scale Green's function. The structure of the fine-scale Green's function was studied extensively in [20] and the interested reader is referred to this work for further details.

Isogeometric analysis (IGA) is a computational mechanics technology that constructs single representations of exact geometry that is preserved under mesh refinement and polynomial degree elevation [21]. The basic idea is to use in analysis the basis-function technology used in computational geometric representations. In modern Computer aided design (CAD) systems, Non-uniform rational B-splines (NURBS) are the dominant technology.

The isogeometric residual-based VMS procedure has been applied successfully to the solution of laminar and turbulent flows along the concentric annular cylinders [22, 23]. A single numerical procedure has been shown, for the first time, to exhibit optimal rates of convergence to analytical solutions of laminar flows, LES-quality results on medium-resolution simulations of turbulent flows, and convergence to DNS solutions of turbulent flows with mesh refinement. This represents a very positive result for the methodology utilized and indicates its strong potential for engineering applications in which flows often contain laminar and turbulent subregimes.

In this paper, we apply the proven technology, namely isogeometric residual-based VMS procedure, to the flow through eccentric annuli. Through the Isogeometric representation of the eccentric annuli, the exact geometry of the circular boundary could be realized and the singular behavior of the solution near the contact point between the two eccentric circular cylinders was analyzed precisely.

This paper is organized as follows. In Sec. 2 the strong and weak forms of the incompressible Navier-Stokes equations and the discrete VMS formulation are presented. In Sec. 3 our general approach to build an exact geometric model is presented. In Sec. 4 we present our numerical studies for laminar flow. In Sec. 5 we summarize our observations and present our conclusions.

2. Variational multiscale method for incompressible Navier-Stokes equations

The variational multiscale method is a procedure for deriving models and numerical methods capable of dealing with multiscale phenomena ubiquitous in science and engineering. It is motivated by the simple fact that straightforward application of Galerkin's method employing standard bases, such as Fourier series and finite elements, is not a robust.

2.1 Strong and weak formulations

Let us start by recalling the incompressible Navier-Stokes equations. Let $\Omega \subset \mathbb{R}^d, d = 2,3$, denote the spatial domain occupied by the fluid, and let $\Gamma = \partial\Omega$ be its boundary. Then

$$\frac{\partial \mathbf{u}}{\partial \mathrm{t}} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) + \nabla p =$$

$$\nabla \cdot \left(2\nu \nabla^s \mathbf{u}\right) + \mathbf{f} \quad \text{in } \Omega, t \in]0, T[, \tag{1}$$

$$\nabla \cdot \mathbf{u} = 0 \qquad \text{in } \Omega, t \in]0, T[, \tag{2}$$

where $\mathbf{u}$ is the velocity, $\mathbf{f}$ is the body force (per unit mass), ν is the kinematic viscosity, p is the pressure divided by the density and $\otimes$ denotes the tensor product (e.g., in component notation, $[u \otimes v]_{ij} = u_i v_j$). Eqs. (1) and (2) are the balance of linear momentum and incompressibility constraint. These equations must be supplied with an initial condition of the form $\mathbf{u} = \mathbf{u}_0$ in Ω, $t = 0$ and a boundary condition which, for simplicity, will be taken as $\mathbf{u} = g$ on $t \in]\ T[$. We also define $\nabla^s \mathbf{u}$ as follows:

$$\nabla^{\mathrm{s}} \mathbf{u} = \frac{1}{2}\left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right). \tag{3}$$

To formulate the weak statement of the problem, let $\mathcal{V}$ denote both the trial solution and weighting function spaces, which are assumed to be identical. We assume $\mathbf{U} = \{\mathbf{u}, p\} \in \mathcal{V}$ implies $\mathbf{u} = g$ on Γ for all $t \in]0, T[$. The variational formulation is stated follows: Find $\mathbf{U} \in \mathcal{V}$ such that $\forall \mathbf{W} = \{\mathbf{w}, q\} \in \mathcal{V}$

$$B(\mathbf{W}, \mathbf{U}) = L(\mathbf{W}) \tag{4}$$

where

$$B(\mathbf{W}, \mathbf{U}) = \left(\mathbf{w}, \frac{\partial \mathbf{u}}{\partial t}\right)\Omega - (\nabla \mathbf{w}, \mathbf{u} \otimes \mathbf{u})_\Omega + (q, \nabla \cdot \mathbf{u})_\Omega$$
$$-(\nabla \cdot \mathbf{w}, p)_\Omega + \left(\nabla^s \mathbf{w}, 2\nu \nabla^s \mathbf{u}\right)_\Omega \tag{5}$$

$$L(\mathbf{W}) = (\mathbf{w}, \mathbf{f})_\Omega . \tag{6}$$

2.2 Residual-based variational multiscale method

We consider a direct-sum decomposition of $\mathcal{V}$ into coarse-scale and fine-scale subspace, $\mathcal{V}^h$ and $\mathcal{V}'$ respectively,

$$\mathcal{V} = \mathcal{V}^h \oplus \mathcal{V}'. \tag{7}$$

$\mathcal{V}^h$ is assumed to be a finite-dimensional space, while $\mathcal{V}'$ is infinite-dimensional.

The first step consists of the multiscale decomposition of the original fields,

$$B(\mathbf{W}^h, \mathbf{U}^h + \mathbf{U}') = L(\mathbf{W}^h) \tag{8}$$

$$B(\mathbf{W}', \mathbf{U}^h + \mathbf{U}') = L(\mathbf{W}') \tag{9}$$

where $\mathbf{U}^h = \{\mathbf{u}^h, p^h\}$ and $\mathbf{U}' = \{\mathbf{u}', p'\}$ stand for the coarse-scale (Resolved-scale) and fine-scale (Unresolved-scale) components of the solution, respectively. Eqs. (8) and (9) are the coarse- and fine-scale equations, respectively. The left-hand side of Eq. (8) consists of the following terms:

$$\begin{aligned} B(\mathbf{W}^h, \mathbf{U}^h + \mathbf{U}') = & \left(\mathbf{w}^h, \frac{\partial \mathbf{u}^h}{\partial t}\right)_\Omega + \left(\mathbf{w}^h, \frac{\partial \mathbf{u}'}{\partial t}\right)_\Omega \\ & - \left(\nabla \mathbf{w}^h, \mathbf{u}^h \otimes \mathbf{u}^h\right)_\Omega - \left(\nabla \mathbf{w}^h, \mathbf{u}^h \otimes \mathbf{u}'\right)_\Omega \\ & - \left(\nabla \mathbf{w}', \mathbf{u}^h \otimes \mathbf{u}^h\right)_\Omega - \left(\nabla \mathbf{w}^h, \mathbf{u}' \otimes \mathbf{u}'\right)_\Omega \\ & + \left(q^h, \nabla \cdot \mathbf{u}^h\right)_\Omega + \left(q^h, \nabla \cdot \mathbf{u}'\right)_\Omega \\ & - \left(\nabla \cdot \mathbf{w}^h, p^h\right)_\Omega - \left(\nabla \cdot \mathbf{w}^h, p'\right)_\Omega \\ & + \left(\nabla^s \mathbf{w}^h, 2\nu \nabla^s \mathbf{u}^h\right)_\Omega + \left(\nabla^s \mathbf{w}^h, 2\nu \nabla^s \mathbf{u}'\right)_\Omega . \end{aligned} \tag{10}$$

We can rewrite Eq. (10) as follows:

$$\begin{aligned} B(\mathbf{W}^h, \mathbf{U}^h + \mathbf{U}') = & B(\mathbf{W}^h, \mathbf{U}^h) + \left(\mathbf{w}^h, \frac{\partial \mathbf{u}'}{\partial t}\right)_\Omega \\ & - \left(\nabla \mathbf{w}^h, \mathbf{u}^h \otimes \mathbf{u}'\right)_\Omega - \left(\nabla \mathbf{w}', \mathbf{u}^h \otimes \mathbf{u}^h\right)_\Omega \\ & - \left(\nabla \mathbf{w}^h, \mathbf{u}' \otimes \mathbf{u}'\right)_\Omega + \left(q^h, \nabla \cdot \mathbf{u}'\right)_\Omega \\ & - \left(\nabla \cdot \mathbf{w}^h, p'\right)_\Omega + \left(\nabla^s \mathbf{w}^h, 2\nu \nabla^s \mathbf{u}'\right)_\Omega \end{aligned} \tag{11}$$

where

$$\begin{aligned} B(\mathbf{W}^h, \mathbf{U}^h) = & \left(\mathbf{w}^h, \frac{\partial \mathbf{u}^h}{\partial t}\right)_\Omega - \left(\nabla \mathbf{w}^h, \mathbf{u}^h \otimes \mathbf{u}^h\right)_\Omega \\ & + \left(q^h, \nabla \cdot \mathbf{u}^h\right)_\Omega - \left(\nabla \cdot \mathbf{w}^h, p^h\right)_\Omega \\ & + \left(\nabla^s \mathbf{w}^h, 2\nu \nabla^s \mathbf{u}^h\right)_\Omega . \end{aligned} \tag{12}$$

The first term on the right-hand side of Eq. (11) is referred to as the Galerkin term, which is defined in Eq. (12); the second term is assumed to be equal to zero because the time derivative of $\mathbf{u}'$ is neglected, leading to a quasi-static modeling of the fine-scales. The third and fourth terms represent cross stress and the fifth term is the Reynolds stress. The fourth and fifth terms produced by the variational multiscale method are not accounted for in classical stabilization methods, such as SUPG and GLS, which only include the third term. The last term on the right-hand side of Eq. (11) is assumed to be zero due to an orthogonality condition induced by the Dirichlet projector. See Ref. [17] for further details and elaboration.

Based on a careful observation of the fine-scale Eq. (9), the fine scale solution can be modeled using the residual of the coarse-scale solution. Eq. (13) illustrates the fact that the fine scales are "driven" by the residual of the coarse-scale equation, $\mathbf{r}_M$. In addition to the above simplifying assumptions, $\mathbf{u}'$ is approximated through an algebraic model. We model the fine scales as in Ref. [24]:

$$\mathbf{U}' \approx -\boldsymbol{\tau} \mathbf{R}(\mathbf{U}^h) \tag{13}$$

where τ is a 4×4 matrix and $\mathbf{R}(\mathbf{U}^h)$ is a 4×1 vector that collects momentum and continuity residual of the Navier-Stokes equations,

$$\mathbf{R}(\mathbf{U}^h) = \left\{\mathbf{r}_M^T(\mathbf{u}^h, p^h), r_C(\mathbf{u}^h)\right\}^T . \tag{14}$$

We define $\boldsymbol{\tau}$ as follows:

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_M I_{3\times 3} & \mathbf{O}_3 \\ \mathbf{O}_3^T & \tau_C \end{bmatrix} \tag{15}$$

where

$$\tau_M = \left(\frac{C_t}{\Delta t^2} + \mathbf{u}^h \cdot \mathbf{G}\mathbf{u}^h + C_I \nu^2 \mathbf{G} : \mathbf{G}\right)^{-1/2} \tag{16}$$

$$\tau_C = (\mathbf{g} \cdot \tau_M \mathbf{g})^{-1} \tag{17}$$

with $\mathbf{G}$ a second rank metric tensor

$$\mathbf{G} = \frac{\partial \boldsymbol{\xi}^T}{\partial \mathbf{x}} \frac{\partial \boldsymbol{\xi}}{\partial \mathbf{x}} \tag{18}$$

and $\mathbf{g}$ a vector obtained from the column sums of $\frac{\partial \boldsymbol{\xi}}{\partial \mathbf{x}}$, $\mathbf{g} = \{g_i\}$

$$g_i = \sum_{j=1}^{3} \left(\frac{\partial \xi_i}{\partial x_j} \right). \tag{19}$$

x and ξ denote the coordinates of elements in physical and parametric space, respectively. Also, Δt is the time step size and C_t and C_I are non-dimensional positive constants, independent of the mesh size. C_t is set to 4 and C_I is considered 36, 36×4 and 36×9 for linear, quadratic and cubic elements, respectively.

Combining Eqs. (8), (11) and (14), we obtain our discrete formulation: Find $\mathbf{U}^h$ such that $\forall \mathbf{W}^h$

$$B^{MS}\left(\mathbf{W}^h, \mathbf{U}^h\right) = L^{MS}\left(\mathbf{W}^h\right) \tag{20}$$

where

$$\begin{aligned} B^{MS}\left(\mathbf{W}^h, \mathbf{U}^h\right) &= B^{G}\left(\mathbf{W}^h, \mathbf{U}^h\right) \\ &+ \left(\mathbf{u}^h \cdot \nabla \mathbf{w}^h + \nabla q^h, \tau_M \mathbf{r}_M\right)_\Omega \\ &+ \left(\nabla \cdot \mathbf{w}^h, \tau_C r_C\right)_\Omega + \left(\mathbf{u}^h \cdot \left(\nabla \mathbf{w}^h\right)^T, \tau_M \mathbf{r}_M\right)_\Omega \\ &- \left(\nabla \mathbf{w}^h, \tau_M \mathbf{r}_M \otimes \tau_M \mathbf{r}_M\right)_\Omega \end{aligned} \tag{21}$$

$$L^{MS}\left(\mathbf{W}^h\right) = \left(\mathbf{w}^h, \mathbf{f}\right)_\Omega \tag{22}$$

and

$$\begin{aligned} B^{G}\left(\mathbf{W}^h, \mathbf{U}^h\right) &= \left(\mathbf{w}^h, \frac{\partial \mathbf{u}^h}{\partial t}\right)_\Omega + \left(\nabla^s \mathbf{w}^h, 2\nu \nabla^s \mathbf{u}^h\right)_\Omega \\ &- \left(\nabla \mathbf{w}^h, \mathbf{u}^h \otimes \mathbf{u}^h\right)_\Omega + \left(q^h, \nabla \cdot \mathbf{u}^h\right)_\Omega \\ &- \left(\nabla \cdot \mathbf{w}^h, p^h\right)_\Omega . \end{aligned} \tag{23}$$

The superscripts MS and G stand for multiscale and Galerkin, respectively. Now, we consider the roles of the different terms in Eq. (22). The first term on the right-hand side of Eq. (21), defined in Eq. (23), is the Galerkin term; the next two terms are classical stabilization terms; and the last two terms are the additional terms produced by the variational multiscale method. From this perspective, classical stabilization methods, such as SUPG and GLS (see Ref. [16]), are viewed as only stepping stones toward the full variational multiscale method.

In the following part, A is the nodal index in standard finite element analysis, and the control point index in NURBS based isogeometric analysis, and e_i is the i th Cartesian basis vector. We assume that velocity and pressure are applied in terms of the same basis, denoted $\{N_A\}_{A=1}^{n_b}$, where n_b is the number of basis functions. Let V, $\dot{V}$, and P denote the vectors of nodal or control point degrees of freedom of velocity, velocity time derivative, and pressure, respectively. We define two residual vectors, corresponding to the momentum and continuity equations, by substituting N_{Ae_i} and N_A in place of w^h and q^h in Eq. (20), respectively,

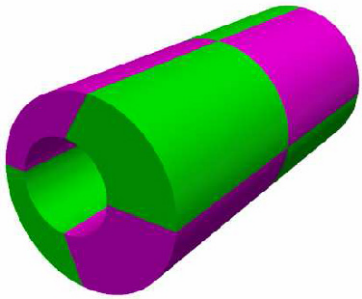


Fig. 1. Exact geometric model of flow domain between eccentric circular cylinders with eccentricity $ec = 0.5$ generated from quadratic NURBS elements.

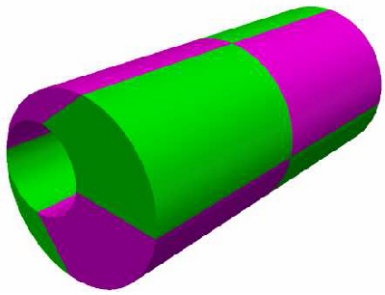


Fig. 2. Exact geometric model of flow domain between eccentric circular cylinders with eccentricity $ec = 1.0$ generated from quadratic NURBS elements.

$$\mathbf{R}^M = \left[R^M_{A,i}\right] \tag{24}$$

$$R^M_{A,i} = B^{MS}\left(\left\{N_{Ae_i}, 0\right\}, \left\{\mathbf{u}^h, p^h\right\}\right) - L^{MS}\left(\left\{N_{Ae_i}, 0\right\}\right) \tag{25}$$

$$\mathbf{R}^C = \left[R^C_A\right]. \tag{26}$$

The generalized $-\alpha$ method [25, 26] is employed to integrate the governing equations in time. This leads to a nonlinear system of equations to be solved at each time step for which we employ a Newton-Raphson procedure with a two-stage predictor-multicorrector algorithm (see Ref. [17] for further details).

3. Geometric model construction and basis functions for analysis

We consider the computational domain with multiple NURBS patches therefore the ducts are divided into 8 patches, as shown in Figs. 1 and 2. The ducts shown in Figs. 1 and 2 has an inner radius of $R_i = 2$, and an outer radius of $R_o = 4$. The length of the cylinder is $L = 18$. Also the geometric models have been constructed based on quadratic NURBS

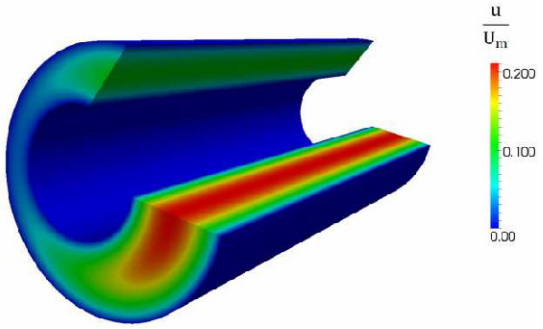


Fig. 3. Laminar flow, snapshot of velocity contours for $\mathrm{Re}=0.2$, based on bulk-flow Reynolds number , for the eccentric duct with eccentricity, $ec=0.5$.

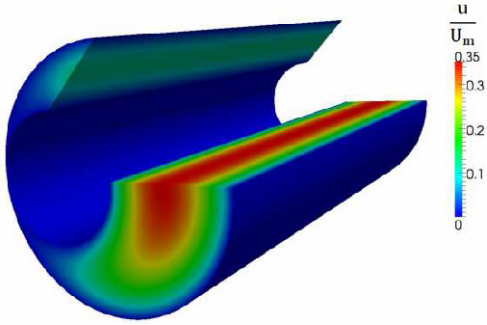


Fig. 4. Laminar flow, snapshot of velocity contours for $\mathrm{Re}=0.2$, based on bulk-flow Reynolds number , for the eccentric duct with eccentricity, $ec=1.0$.

basis functions in the all directions. By using the quadratic NURBS functions for the geometry representation, the circular geometry can be represented exactly and this is the major difference from the classical finite element/volume method. More detail of the NURBS based geometrical modeling and analysis can be found in the Ref. [21].

4. Laminar flow

This section describes a laminar verification of the numerical formulation for the duct with eccentricity, $\mathrm{ec}=0.5$ and introduces a laminar numerical result for the duct which has eccentricity, $\mathrm{ec}=1.0$. The flow is chosen as a test case for the duct with eccentricity, $\mathrm{ec}=0.5$, because the exact solution, whose detail is presented in the Appendix A, is available to compare with the numerical results. Mesh refinement studies are performed for each order of solution approximation utilized, namely linear and quadratic basis functions.

4.1 Problem setup

The flow domains are illustrated in Figs. 3 and 4. A no slip Dirichlet boundary condition is set at the walls. The flow is driven by a constant pressure gradient, f_x, acting in the stream-wise direction. The values of the kinematic viscosity

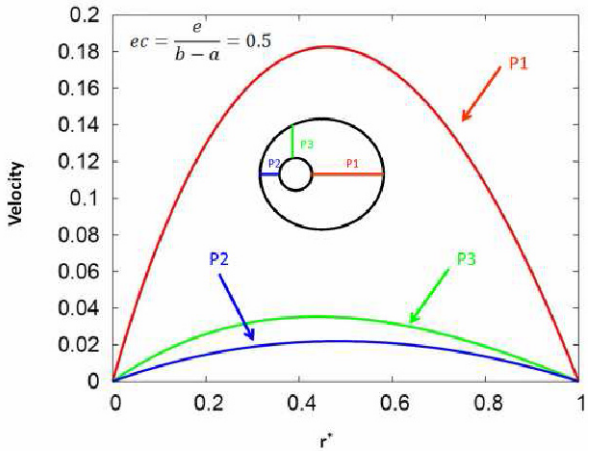


Fig. 5. Velocity profiles of the eccentric duct with eccentricity, $ec=0.5$ on the different planes at $\mathrm{Re}=0.2$, based on bulk-flow Reynolds number.

ν and the forcing f_x are set to 10 and 2.0, respectively.

Computational cost for the current simulation is minimal compared to the turbulent flow simulation, because the cases considered for the current papers are all low Reynolds number problems, and the simulation reaches the steady-state immediately. The major difference of the current method compared to the classical finite volume method would be element-wise integration step for the construction of the system of algebraic equations, which a linear algebra solver eventually has to solve. Except those step, the solution procedures are pretty much same irrespective of the discretization methods.

4.2 Numerical results

4.2.1 Eccentric duct with eccentricity, $ec=0.5$

The computations were performed on 16^3, 32^3 and 64^3 elements. We employ C^0-continuous linear finite elements and C^1-continuous quadratic NURBS. For all orders, the number of basis functions is equal to the number of elements in the stream-wise and circumferential directions. In the wall-normal direction, due to the open knot vector construction, the number of basis functions is $n+p$, where n is the number of elements in the wall-normal direction and p is the polynomial order. Fig. 5 shows velocity profiles on different planes.

Our numerical results are compared with the analytical solution. The rates of convergence of the error measured in the L^2-norm of velocity versus mesh parameter, are presented in Fig. 6. As may be seen in Fig. 6, the slopes of the lines related to linear and quadratic approximations are approximately two and three, indicating second- and third-order accuracy, respectively. These results represent optimal rate of convergence in each case. Distribution of the wall shear stress τ along the inner and outer walls is shown in Fig. 7.

4.2.2 Eccentric duct with eccentricity, $ec=1.0$

The computations were performed on 16^3, 32^3 and 64^3

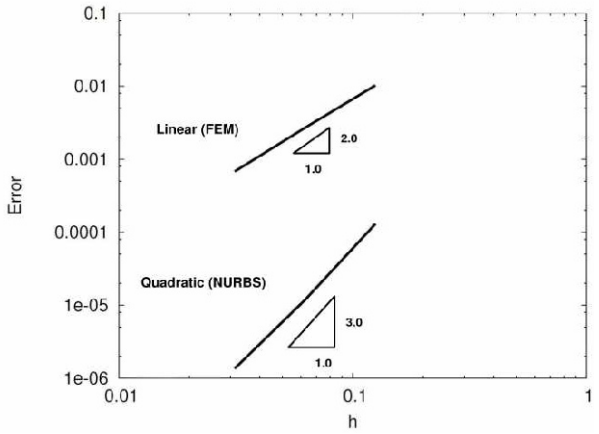


Fig. 6. Convergence of the error in the L^2 -norm of the velocity for linear finite elements, and quadratic and cubic NURBS discretization.

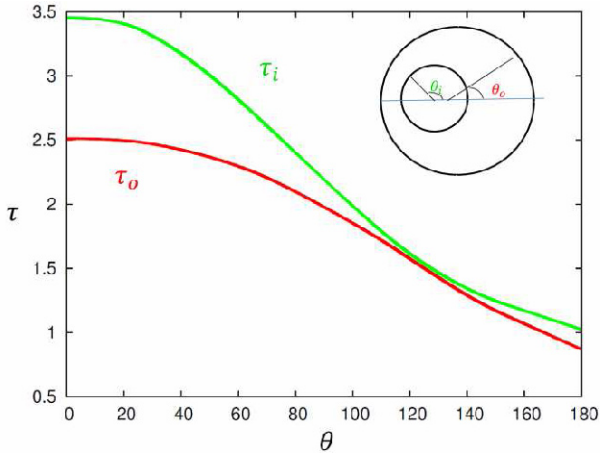


Fig. 7. Distribution of the wall shear stress τ along the inner and outer walls.

elements. We employ C^1 -continuous quadratic NURBS. For all orders, the number of basis functions is equal to the number of elements in the stream-wise and circumferential directions. In the wall-normal direction, due to the open knot vector construction, the number of basis functions is $n+p$, where n is the number of elements in the wall-normal direction and p is the polynomial order. Fig. 8 shows velocity profile on the plane located at the widest gap. Distribution of the wall shear stress τ along the inner and outer walls is shown in Fig. 9.

5. Summary and conclusions

In this paper we have studied the behavior of the Variational multiscale (VMS) method and Isogeometric analysis (IGA) in the computation of laminar eccentric annular channel flows governed by the incompressible Navier-Stokes equations. In the laminar case, rapid, and in fact optimal, rates of convergence for linear finite elements and quadratic NURBS were observed. NURBS of quadratic facilitate the construction of exact cylindrical geometric domains. In the laminar regime,

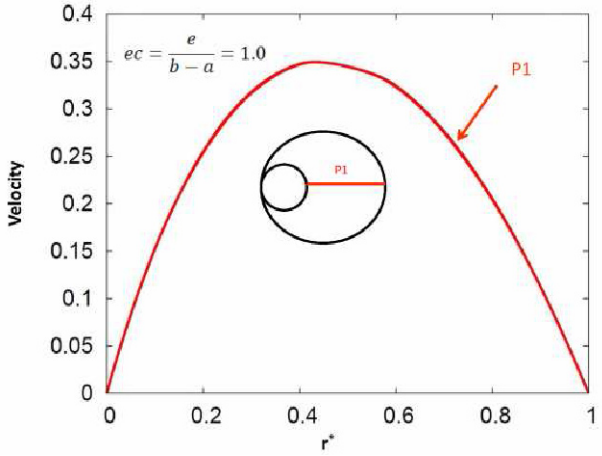


Fig. 8. Velocity profile of the eccentric duct with eccentricity, $ec = 1.0$ on the plane located at the widest gap at $\mathrm{Re} = 0.2$, based on bulk-flow Reynolds number.

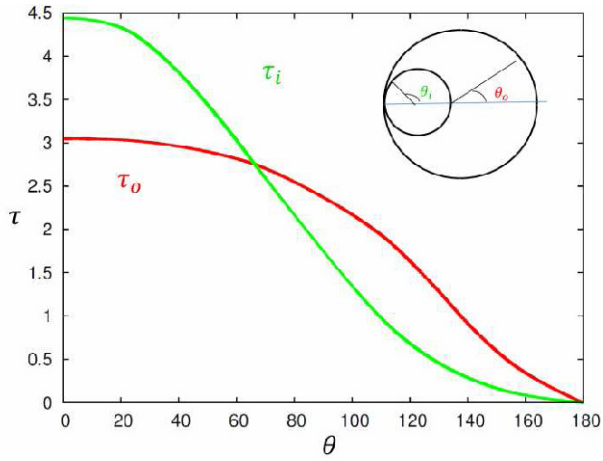


Fig. 9. Distribution of the wall shear stress τ along the inner and outer walls.

VMS behaves like a higher-order accurate, residual-driven stabilized method. The utility of the methodology for solving flows in an annular region in which the inner cylinder touches the outer cylinder, creating a singularity, has been demonstrated. There are no other analytical or numerical results for this case. Analytical methods fail because bipolar coordinates degenerate for touching cylinders, and traditional numerical methods fail because mesh zones become concave. Only isogeometric methods have been shown to provide optimal rates of convergence for concave mesh zones.

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Nomenclature

u : Velocity

p : Pressure divided by the density
ν : Kinetic viscosity
f : Body force per unit mass
t : Time

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Appendix

A. An analysis of fully developed laminar flow between two eccentric cylinders

An analysis, developed for the fully developed laminar flow in an eccentric annulus by Snyder et al. [6, 7], is presented to validated the present our numerical method. By the analytical method based on the bipolar transformation, an exact solution

for the velocity distribution is calculated and then compare with that obtained by the numerical method shown in Sec. 4.

$$u(\xi,\eta) = E_0 + E_{1\eta} + \sum_{n=1}^{\infty}\left[A_n e^{n\eta} + B_n e^{-n\eta}\right]\cos n\xi + \frac{Qc^2}{a^2}\left[1 - \frac{2\cos\xi}{\cosh\eta + \cos\xi}\right] \quad (A.1)$$

where

$$E_0 = \frac{Qc^2}{a^2}\left[1 - 2\frac{\alpha\coth\beta - \beta\coth\alpha}{\alpha - \beta}\right]$$

$$E_1 = \frac{2Qc^2}{a^2}\frac{\coth\beta - \coth\alpha}{\alpha - \beta}$$

$$A_n = \frac{4Qc^2}{a^2}(-1)^n\frac{\coth\beta - \coth\alpha}{e^{2n\alpha} - e^{2n\beta}}$$

$$B_n = \frac{4Qc^2}{a^2}(-1)^n\frac{e^{2n\beta}\coth\alpha - e^{2n\alpha}\coth\beta}{e^{2n\alpha} - e^{2n\beta}}$$

and

$$Q = \frac{a^2}{4\mu}\frac{dP}{dx}.$$

The bipolar coordinate (ξ,η) are defined by the transformation

$$y + iz = c\tan\frac{\xi + i\eta}{2} \quad (A.2)$$

where c is a constant. Thus, one can get relations between the physical and bipolar coordinates in the form

$$y = \frac{c\sin\xi}{\cosh\eta + \cos\xi}, \qquad z = \frac{c \quad \eta}{\cosh\eta + \cos\xi}$$

$$\tan\xi = -\frac{2cy}{y^2 + z^2 - c^2}, \qquad e^{2\eta} = \frac{y^2 + (z+c)^2}{y^2 + (z-c)^2}$$

$$y^2 + \left(z^2 - c\coth\eta\right)^2 = \frac{c^2}{\sinh^2\eta}$$

$$y^2 + \left(z^2 - c\coth\eta\right)^2 = \frac{c^2}{\sinh^2\eta}$$

$$c = a\sinh\alpha = b\sinh\beta$$

$$\cosh\alpha = \frac{b^2 - a^2 - e^2}{2ae} \qquad \beta = \frac{b^2 - a^2 + e^2}{2be}$$

and y, z are related to the cylindrical coordinates r, θ as

$$y = -r\sin\theta, \qquad z = c\coth\alpha - r\cos\theta.$$



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